

Python Practice Problems: Lists, Tuples, Dictionaries, and NumPy

Part 1: Python Lists

Problem 1: List Operations

- 1 Create a list containing the numbers 5, 12, 7, 3, 18, 9.
- 2 Append 15 to the list.
- 3 Insert 20 at index 2.
- 4 Remove the element 7 from the list.
- 5 Sort the list in descending order and print the final list.

Problem 2: List Slicing

- 1 Given numbers = [2,4,6,8,10,12,14,16].
- 2 Extract the first four elements.
- 3 Extract the last three elements.
- 4 Extract elements from index 2 to 5.
- 5 Reverse the list using slicing.
- 6 Extract every second element.

Problem 3: List Processing

- 1 Given scores = [55,78,62,49,85,91,73].
- 2 Create a new list containing only scores greater than 70.
- 3 Calculate the average score.
- 4 Find the maximum and minimum scores.

Part 2: Tuples

Problem 4: Tuple Basics

- 1 Create a tuple containing five programming languages.
- 2 Print the first and last element.
- 3 Check whether 'Python' exists in the tuple.
- 4 Convert the tuple into a list and append 'Rust'.

Problem 5: Tuple Unpacking

- 1 Given student = ('Alice',21,'Computer Science',3.75).
- 2 Unpack the tuple into name, age, major, gpa.

- 3 Print a formatted statement describing the student.

Part 3: Dictionaries

Problem 6: Dictionary Creation

- 1 Create a dictionary containing name, age, department, cgpa.
- 2 Print all keys and values.
- 3 Update the cgpa value.
- 4 Add a new key called 'university'.

Problem 7: Dictionary Iteration

- 1 Given marks = {'Math':85,'Physics':78,'Chemistry':82,'Biology':74}.
- 2 Print all subjects and marks.
- 3 Find the subject with the highest marks.
- 4 Calculate the average marks.

Problem 8: Dictionary Filtering

- 1 Given products = {'Laptop':1200,'Phone':800,'Tablet':450,'Monitor':300,'Keyboard':100}.
- 2 Create a new dictionary containing only products with price greater than 500.

Part 4: NumPy Arrays

Problem 9: Basic NumPy Operations

- 1 Create a NumPy array with values [10,20,30,40,50].
- 2 Multiply each element by 2.
- 3 Find the mean of the array.
- 4 Find the maximum value.

Problem 10: Matrix Creation

- 1 Create a 3x3 NumPy matrix with values from 1 to 9.
- 2 Print the second row.
- 3 Print the third column.
- 4 Compute the sum of all elements.

Problem 11: Array Filtering

- 1 Given arr = np.array([5,12,7,20,3,15,8]).
- 2 Extract elements greater than 10.
- 3 Replace all values less than 10 with 0.

Part 5: Combined Topics Problems

Problem 12: Student Records System

- 1 Given students = [('Alice',85),('Bob',72),('Charlie',90),('David',65),('Eva',88)].
- 2 Convert the data into a dictionary where name is the key.
- 3 Create a list of students who scored above 80.
- 4 Calculate the average score using NumPy.

Problem 13: Inventory Management

- 1 Given inventory = {'apple':[10,1.5], 'banana':[25,0.8], 'orange':[15,1.2], 'mango':[8,2.0]}.
- 2 Calculate the total value of each item.
- 3 Store results in a new dictionary.
- 4 Use NumPy to compute the average item value.

Problem 14: Sensor Data Analysis

- 1 Given temps = [22,24,19,23,25,18,21].
- 2 Convert the list into a NumPy array.
- 3 Find mean, max, and min temperature.
- 4 Store results in a dictionary.
- 5 Create a tuple containing (max_temp, min_temp).

Problem 15: Course Grades Analyzer

- 1 Given courses = {'Alice':[85,90,78], 'Bob':[70,75,80], 'Charlie':[92,88,95]}.
- 2 Compute each student's average score using NumPy.
- 3 Store averages in a new dictionary.
- 4 Create a list of students whose average > 85.
- 5 Store the top student and their score in a tuple.

Problem 16: Data Transformation Task

- 1 Given data = [('A',[10,20,30]),('B',[5,15,25]),('C',[12,18,24])].
- 2 Convert the structure into a dictionary.
- 3 Convert each value list into a NumPy array.
- 4 Compute the mean of each array.
- 5 Store results in a new dictionary.